

# Analysis of Ultrasonic Vocalizations of Mice for Autism Detection

A reproducible pipeline — and a calibrated account of what pup ultrasonic calls can, and cannot, tell us about an ASD genotype: on known animals, and on animals we have never heard before.

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In collaboration with Dr. Hava M. Golan · Ben-Gurion University — biology lead & data owner

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— WHY THIS PROJECT

Someone I love was diagnosed with autism at the age of **three**.

When her world was shaped around how she experiences it, she **flourished**.

Early recognition is what opens that door — yet today it rests on slow, subjective, behavioral assessment.

## — THE PROBLEM

# Autism is diagnosed by **behavior** — **slowly, and late**

- ◆ Diagnosis rests on expert **behavioral assessment** — time-consuming and clinician-bound.
- ◆ It is hardest exactly in **early infancy** — the window where intervention helps most.
- ◆ This motivates an **objective, quantitative, early biomarker** of atypical neurodevelopment.

### THE GAP

A signal that is **naturally produced, recordable,** and present **very early** — before language.

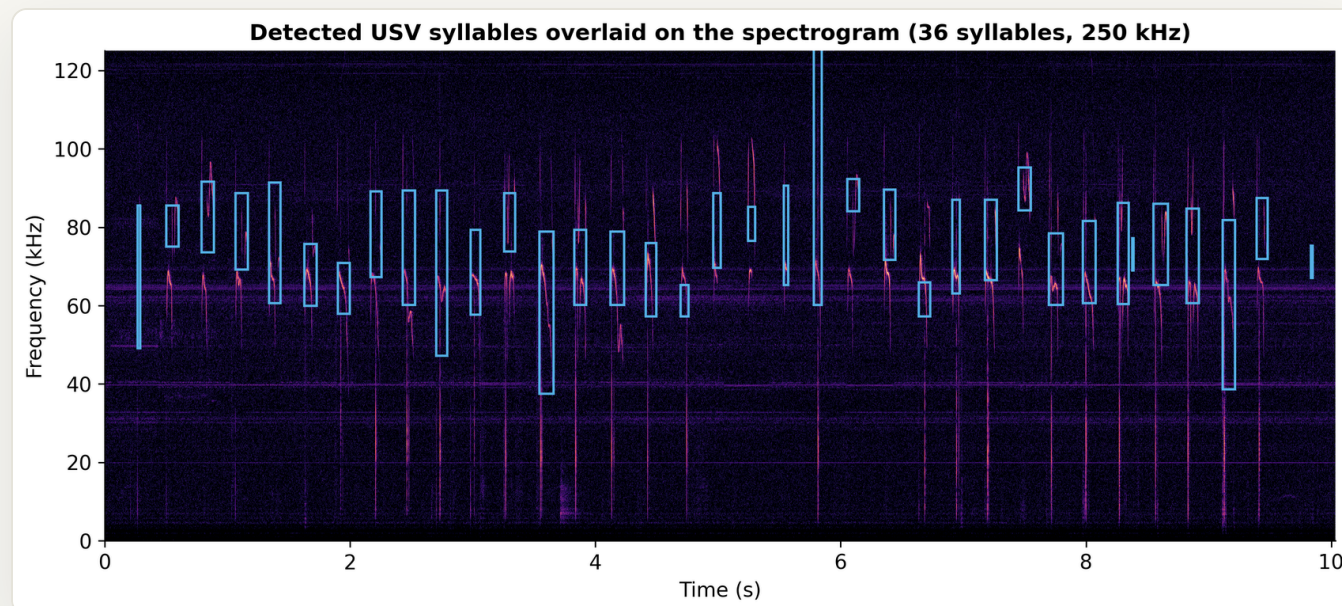
**1 in 36**

children diagnosed with ASD

**< 3y**

where early intervention pays off  
most

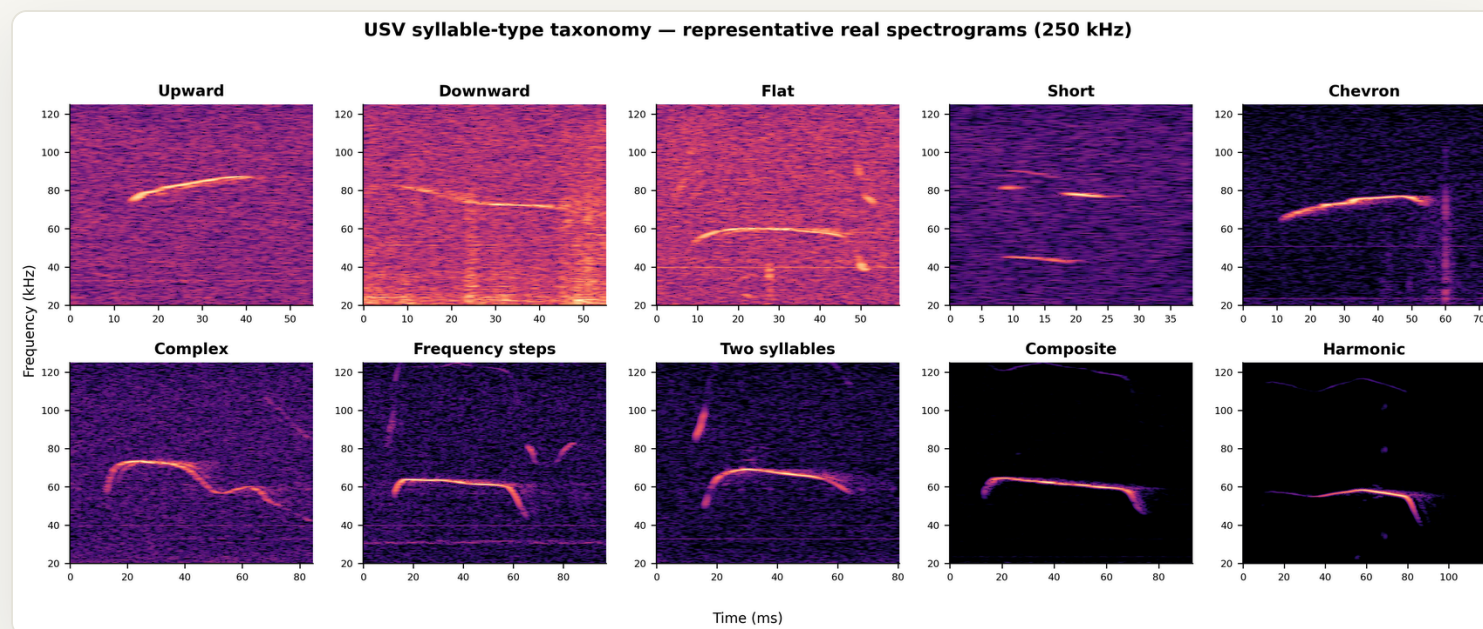
# Mouse pups “speak” — in ultrasound



- ◆ Isolated pups emit **ultrasonic vocalizations** at **35–125 kHz** — inaudible, natural, non-invasive.
- ◆ These isolation calls are among the **earliest communicative behaviors**, sensitive to genotype.
- ◆ Genetic ASD model: **Mthfr**. A pup is **HET** (ASD-model, positive) or **WT** (control).

Each cyan box is one detected **syllable**. Our corpus holds **125,576** of them.

# Calls have shape — and genotype bends it



- ◆ A CNN types each syllable into **10 shapes** — sweeps, steps, harmonic stacks.
- ◆ WT vs HET differ in **start / end frequency** and **duration** (Shekel et al. 2021).

## TWO RESEARCH THREADS

**Acoustic aggregates** per recording · and the **order** of the calls (Gal et al. 2023).



# Four questions — three scientific, one **engineering**

**RQ1** **New models.** Do TabPFN and neural sequence models beat the inherited XGBoost at classifying HET vs WT pups?

**RQ2** **Which features.** Which acoustic characteristics drive classification — do a few consistent ones emerge?

**RQ3** **Regime & generalization.** How does the evaluation split — known vs unseen animals, pooled vs per-strain — change performance?

**ENG** **Reproducibility.** Can a broken, inherited pipeline become a versioned, end-to-end system with a single source of truth?

# We inherited a prototype that didn't run

## INHERITED

### One subject-dependent XGBoost

Hard-coded paths · no documentation · dormant  
~2.5 years · would not run from a clean  
checkout.

## BUILT BY THIS PROJECT

+ Modular, runnable pipeline

+ Subject-independent evaluation

+ Tuned XGBoost

+ TabPFN foundation model

+ 3 sequence models

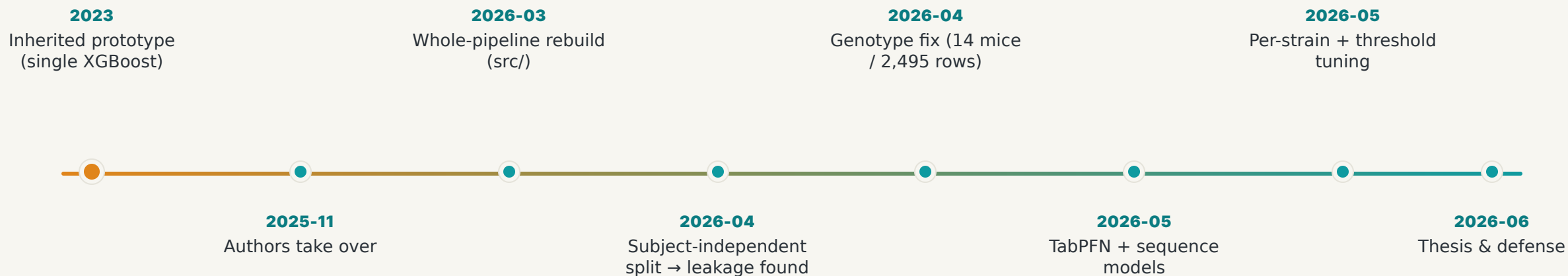
+ Per-strain analysis

+ Single source of truth

+ Graphical curation app

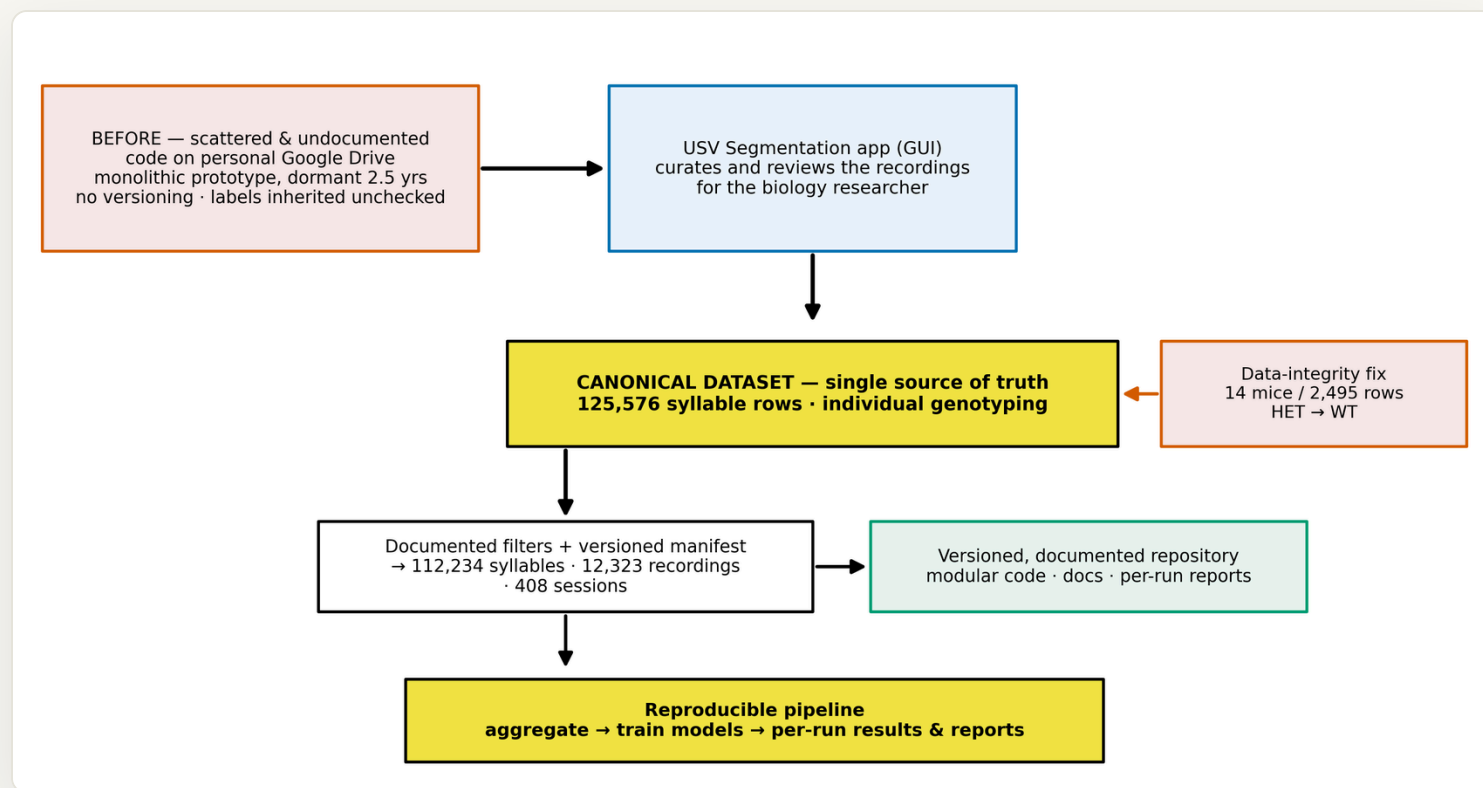
## — THE PROCESS

# From a dead prototype to a reproducible system





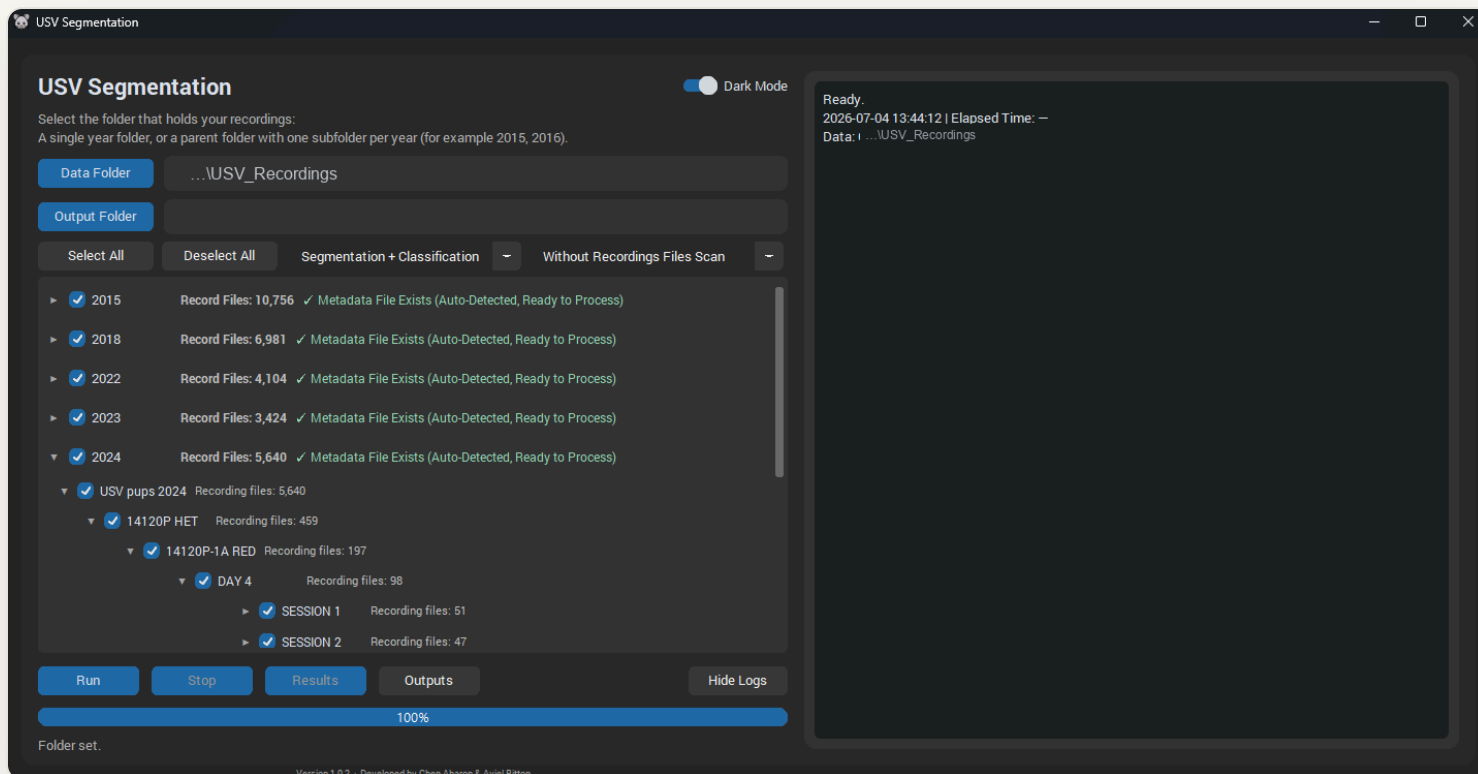
# One source of truth — and a correction it exposed



- ◆ All training now draws from **one canonical file** with a **versioned manifest** — not scattered spreadsheets.
- ◆ Every filter stage and row count is documented and **reproducible**.

Curation exposed a genotype-labelling error: **14 mice · 2,495 rows** corrected **HET → WT**. A HET×WT cross yields ~½ WT pups — genetically necessary, not a regression.

# A tool the biologist actually **uses**

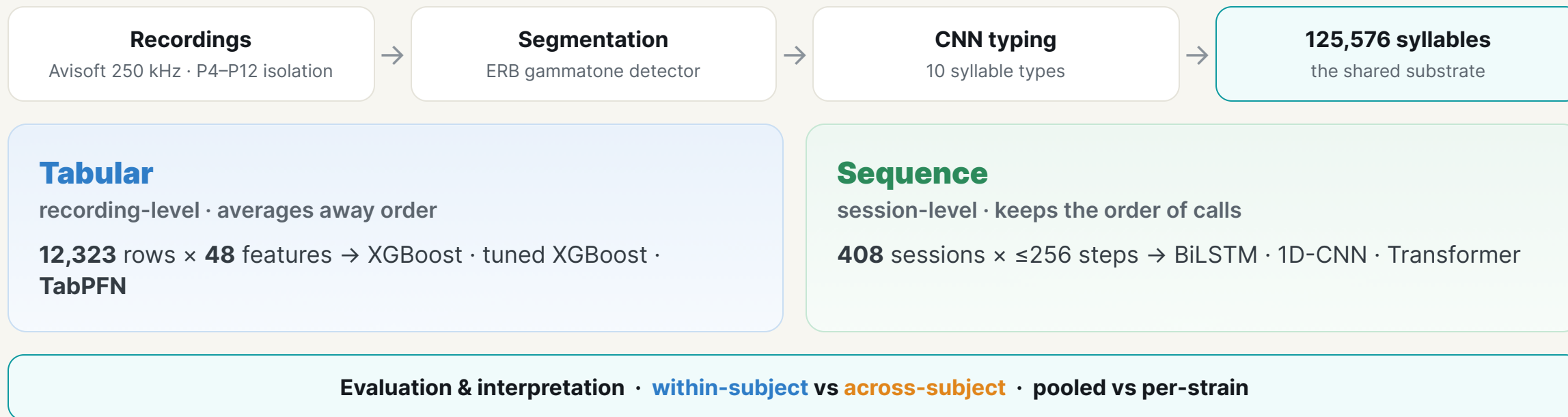


- ◆ A desktop app for **Dr. Hava Golan** — curate & review USVs with **no code**.
- ◆ Wraps the exact **segmentation + CNN typing** pipeline; produces the canonical file all models consume.
- ◆ This is **how the label error was found** — curation in parallel with modelling.

## A REAL, CITABLE ARTIFACT

One-click Windows installer · open source · **Zenodo v1.0.2** (doi:10.5281/zenodo.20096810).

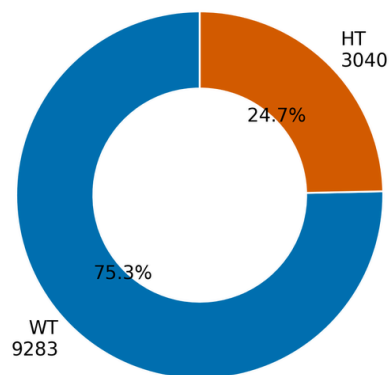
# One recording → two representations → a calibrated call



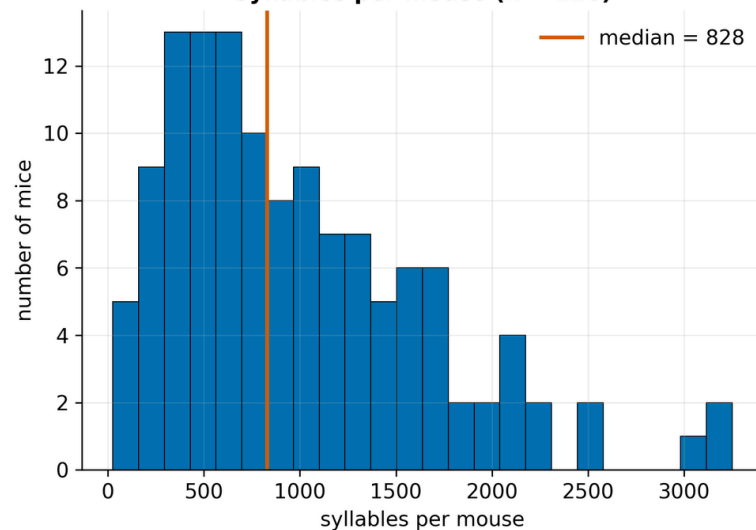
## — THE DATA

# Big, imbalanced, and longitudinal

Recording-level class balance  
(baseline: 12,323 recordings)



Syllables per mouse (n = 126)



126

pups

35

dams

5

years

- ◆ Class balance is **~3:1 WT:HET** — so plain accuracy already scores  $\sim 0.75$  by guessing. We foreground **recall, precision, AUC**.
- ◆ Each pup gives **many correlated recordings** (median 828 syllables) — a naive split leaks the **same animal** into train & test.

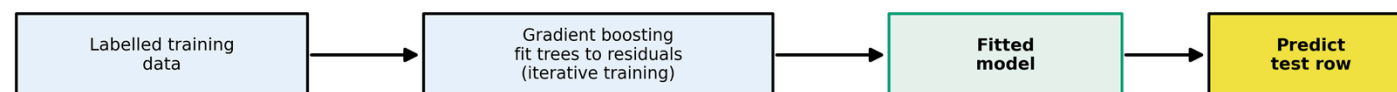
**THIS ONE FACT DRIVES THE WHOLE EVALUATION**

Because animals repeat, we must split **by mouse** — or we fool ourselves.

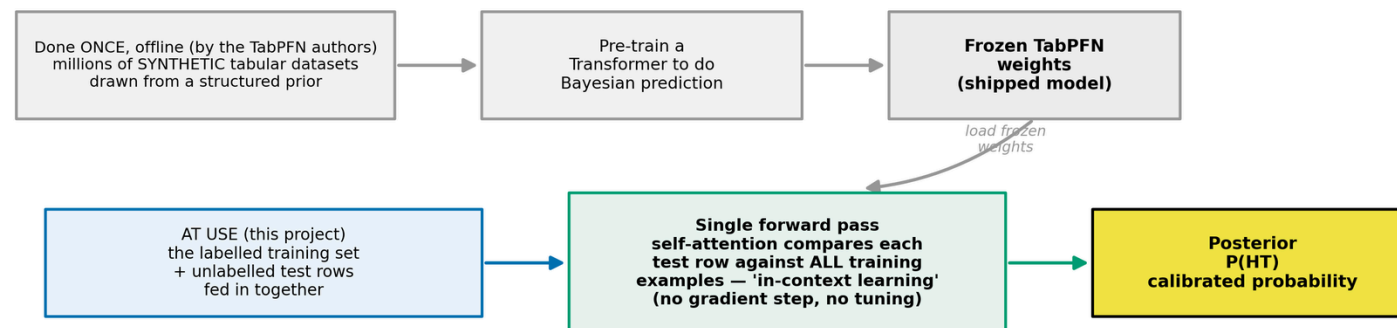
# A tree, a transformer, and no training at all

## Two paradigms for the tabular classifier

### A. Conventional (XGBoost) — trained on the study data



### B. TabPFN — a prior-fitted transformer used in-context (no training on the study data)



Key difference: XGBoost is FITTED on this project's data; TabPFN is FROZEN and run once per prediction, learning the WT-vs-HET boundary from the training rows it is shown in context — fast, tuning-free, suited to the 12,323×48 matrix.

- ◆ **XGBoost** (inherited) — gradient-boosted trees, trained on our data. The reference baseline.
- ◆ **TabPFN** (new) — a transformer **pre-fitted once** on millions of synthetic tables; it does **in-context** inference in a single pass — **no tuning** on our data.

XGBoost · baseline

tuned XGBoost

TabPFN

BiLSTM

1D-CNN

Transformer

# Two questions, reported **side by side**

## Subject-dependent

within-subject · random split

The same animal may appear in train & test. Answers: “**How well on a new recording of a known animal?**”

Best-case, personalized — and directly comparable to prior work.

## Subject-independent

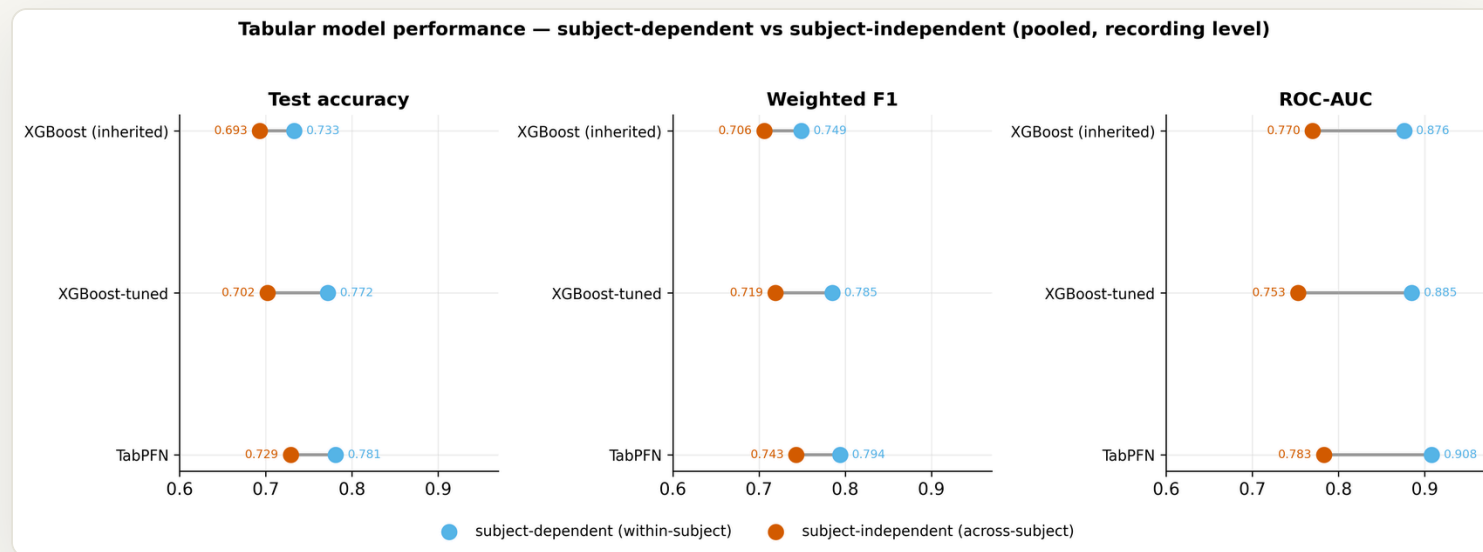
across-subject · grouped by mouse

No animal is shared between sets. Answers: “**How well on an entirely unseen animal?**”

The real generalization test — and it **should** score lower.

Both are legitimate. Reporting **only one** hides half the story — so we always show both.

# TabPFN wins in both regimes



**.781**

KNOWN ANIMALS

**.729**

UNSEEN ANIMALS

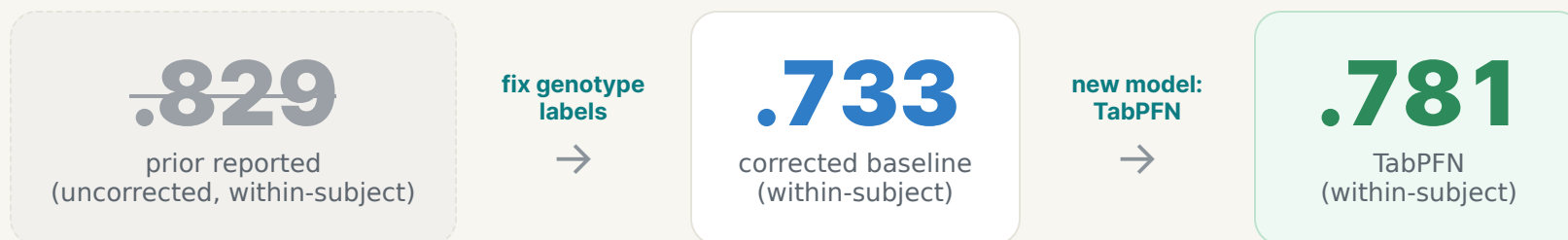
- ◆ Best tabular model on **both** regimes — wF1 .743 · ROC-AUC .783 on unseen mice.
- ◆ Its **unseen**-animal accuracy (.729) ~matches the inherited model on **known** animals (.733).

## CLINICALLY, WHAT MATTERS MOST

Tuned XGBoost keeps **HT recall 0.869** on unseen mice — it rarely **misses** a true ASD-model pup.



# The prior **0.829**, decomposed



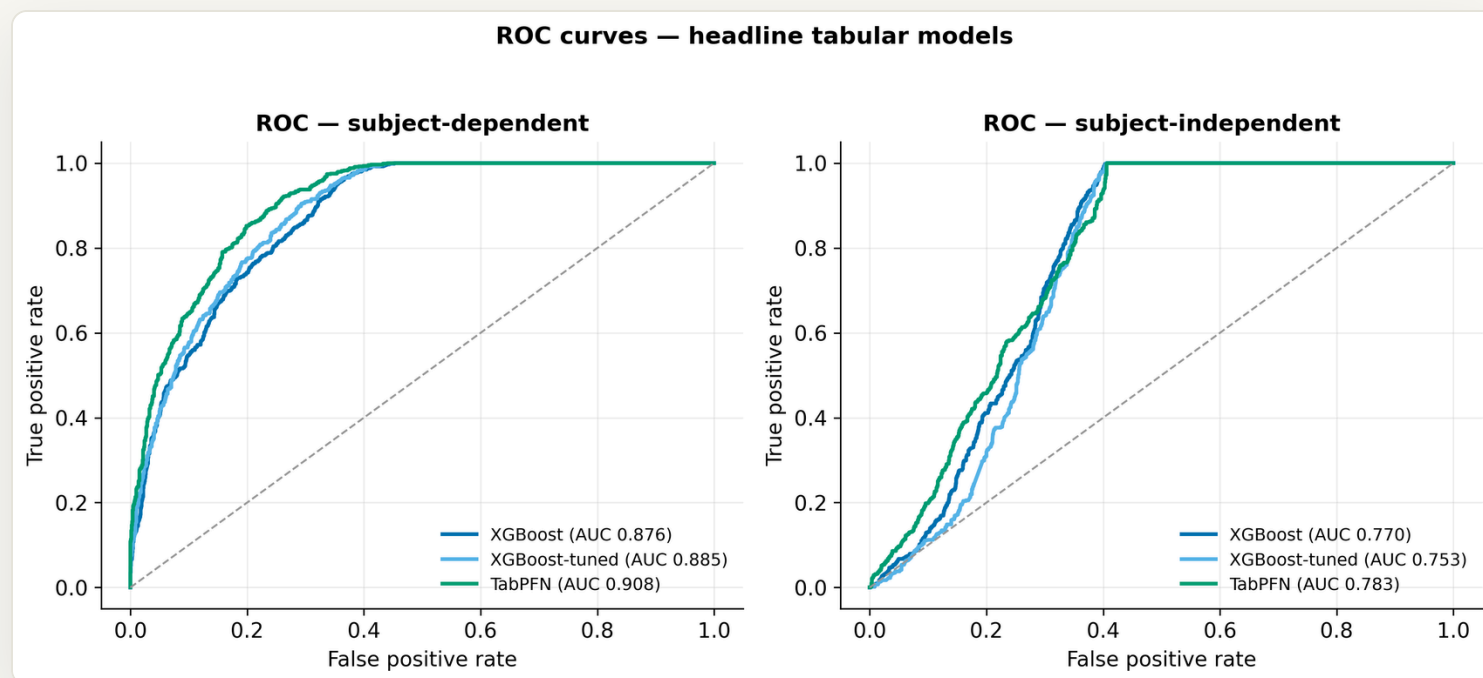
## NOT A REGRESSION — A DATA-INTEGRITY RESULT

The earlier figure was inflated by mislabeled data. **.733** is the genetically-valid baseline — and the new model still **beats** it.

## ON UNSEEN ANIMALS

XGBoost **.693** · TabPFN **.729**  
the expected within→across gap

# The ranking is real — and threshold-free



- ◆ **ROC-AUC** asks: does the model rank a random HET above a random WT? — independent of any threshold.
- ◆ TabPFN leads in both regimes: **.908** known → **.783** unseen.
- ◆ AUC drops from within → across for **every** model — that gap **is** the generalization cost.

## HONEST FOOTNOTE

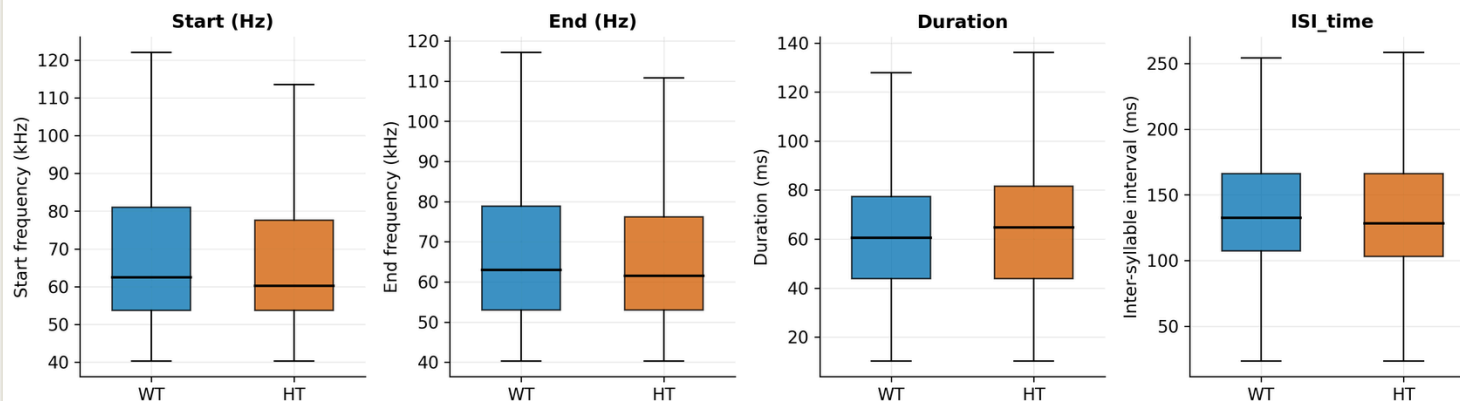
Tuning helped **recall**, not ranking — tuned-XGBoost AUC actually dips below plain XGBoost on unseen mice.

# Why the classes **barely separate** at the call level

~.50

HT PRECISION · EVERY MODEL, EVERY SPLIT

Acoustic-feature distributions by genotype (syllable level)



- ◆ Per-syllable, WT and HET distributions **overlap almost entirely**.
- ◆ So at the default threshold, **~half of predicted-ASD calls are false alarms** — and it survives XGBoost, tuning, and TabPFN alike.

**A FEATURE CEILING, NOT A MODEL LIMIT**

The fix is **richer per-call features** (contour, bandwidth, FM depth) — not a bigger model.

# Does the **order** of the calls carry the signal?

## THE HYPOTHESIS

Gal et al. (2023) found the **temporal dynamics** of isolation calls carry phenotype. So we modelled each session as an **ordered sequence** of syllables ( $\leq 256$  steps, 14 features each) — not just averages.

## Three architectures

BiLSTM (149k) · 1D-CNN (86k) · Transformer (73k) — recurrence, convolution, and attention over the call stream.

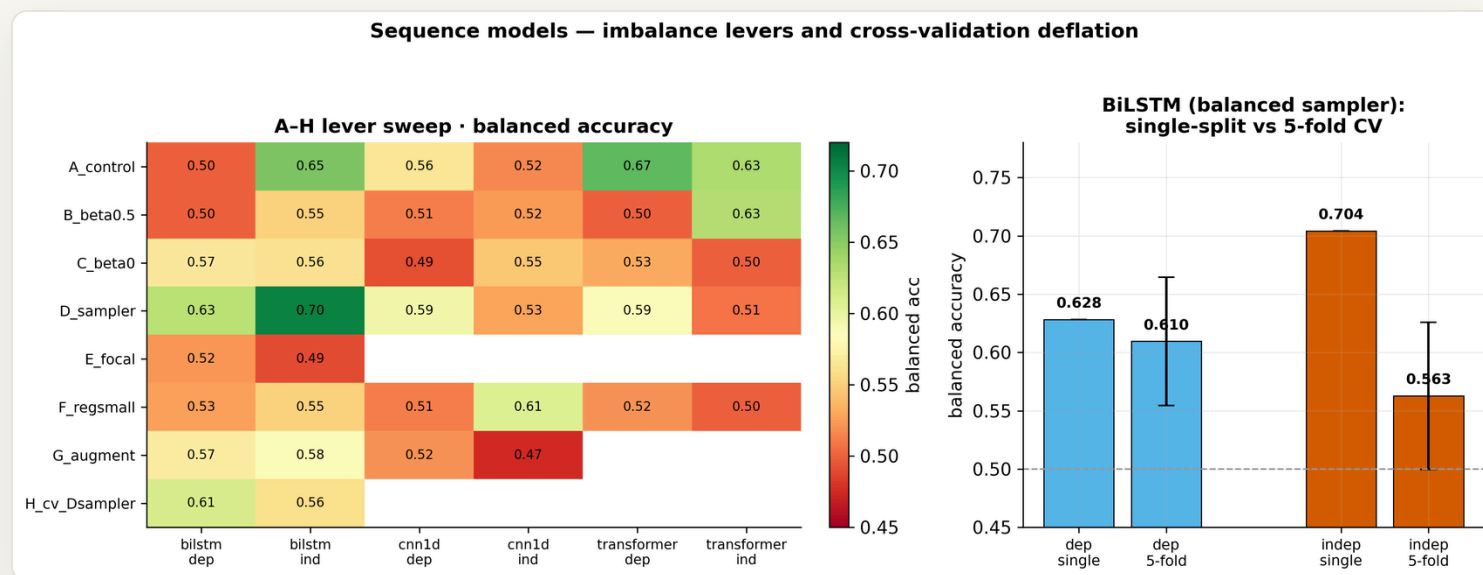
## A systematic search

8 imbalance-handling levers × 3 models × 2 regimes  $\approx$  **36 runs** — loss weighting, balanced sampling, focal loss, augmentation.

## What we hoped

That temporal “syntax” would break the **~0.50 precision wall** the aggregates hit.

# A promising split that **didn't** survive



**.704**  
best single split

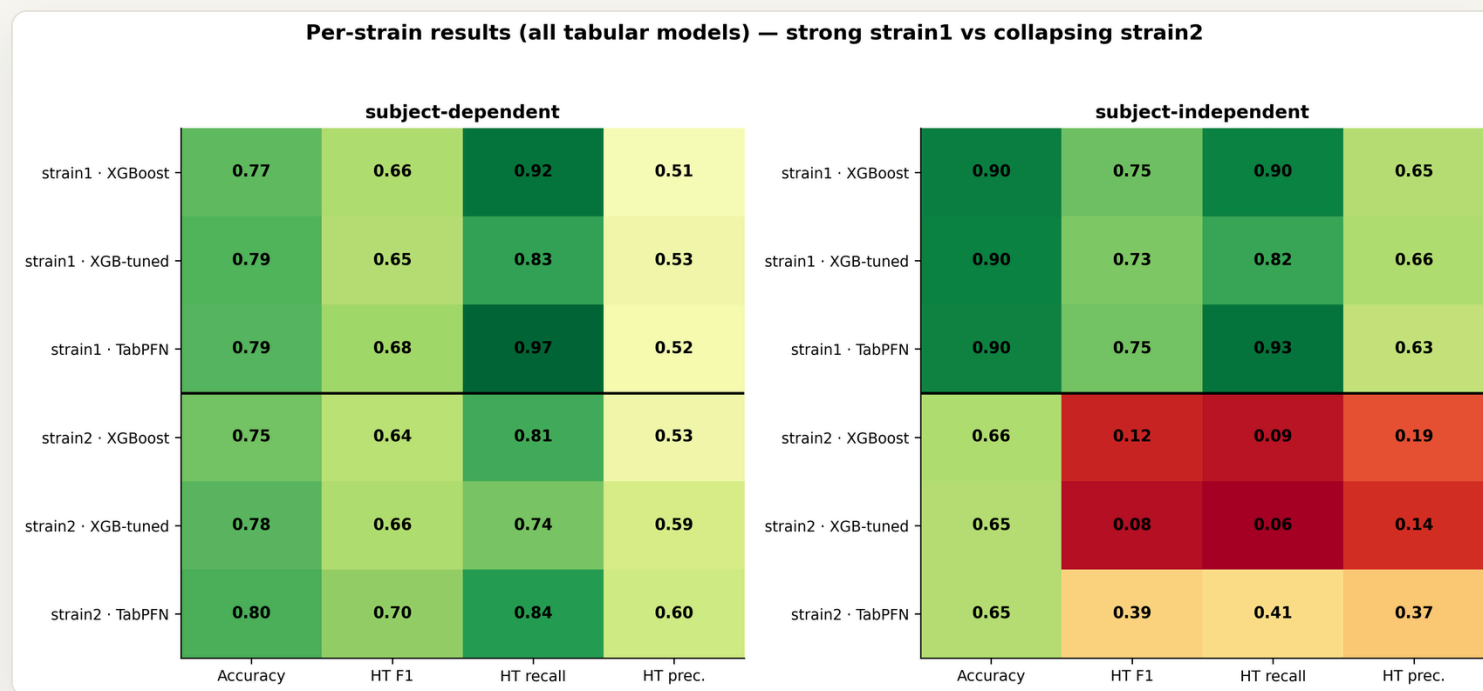
5-fold  
grouped CV  
→

**.563**  
± 0.063

- ◆ The single best run was a **lucky fold**; cross-validation collapses it toward chance.
- ◆ Cause is **data scale** — only ~19 HET test sessions for ~70–150k-parameter nets.

Not a refutation of Gal et al. —  
**“not supported at this data scale.”**  
Order may still help with more animals.

# When 0.90 is a **warning**, not a headline



- ◆ **Strain1** (2022–24) is ~**0.90** across-subject — but it's a newer, more separable cohort.
- ◆ Strain / year / prevalence all **covary with genotype** — the model may read **cohort**, not ASD.
- ◆ On **strain2** (47 mice) the minority class **collapses** — accuracy hides it.

## SO WHICH NUMBER DO WE TRUST?

The **pooled unseen** result — it averages the confound out. The clean test (train one strain, test the other) is future work.

## — CONCLUSIONS

# What USVs **can** — and cannot — support

RQ1 ✓

TabPFN **beats** the baseline in both regimes (.781 / .729); tuned-XGBoost holds **HT recall 0.869** on unseen mice.

RQ2 ✓

Boundary **frequencies & durations** carry the signal (Shekel 2021) — but minority precision is **capped ~0.50** at the feature level.

RQ3 ✓

Known  $\neq$  unseen (gap up to 0.30 in recall); sequence order didn't help at this scale; per-strain is **confounded**.

ENG ✓

A reproducible, end-to-end pipeline with a single source of truth, a curation app, and a corrected, auditable baseline.

The headline is **calibrated**: a strong **within-subject** classifier ( $\sim 0.78$ – $0.80$ ) and honest, above-chance **generalization** to unseen mice ( $0.729 \cdot \text{AUC } 0.78$ ) — and it is all reproducible.

# Where the **next** gain comes from

- 1 **Richer per-call features** — frequency contour, bandwidth, FM depth — to break the precision wall. Highest leverage.
- 2 **A genuine cross-cohort test** — train on one strain, test on the other — to separate phenotype from cohort.
- 3 **Per-mouse decisions** — aggregate recordings into one vote per animal, the clinically meaningful grain.
- 4 **More data for sequence models** — re-test the temporal-dynamics hypothesis at larger scale.
- 5 **Validate the front end** — segmentation vs ground truth, and a model card for the syllable typer.



— REPRODUCIBLE · HONEST · PUBLIC

# Thank you

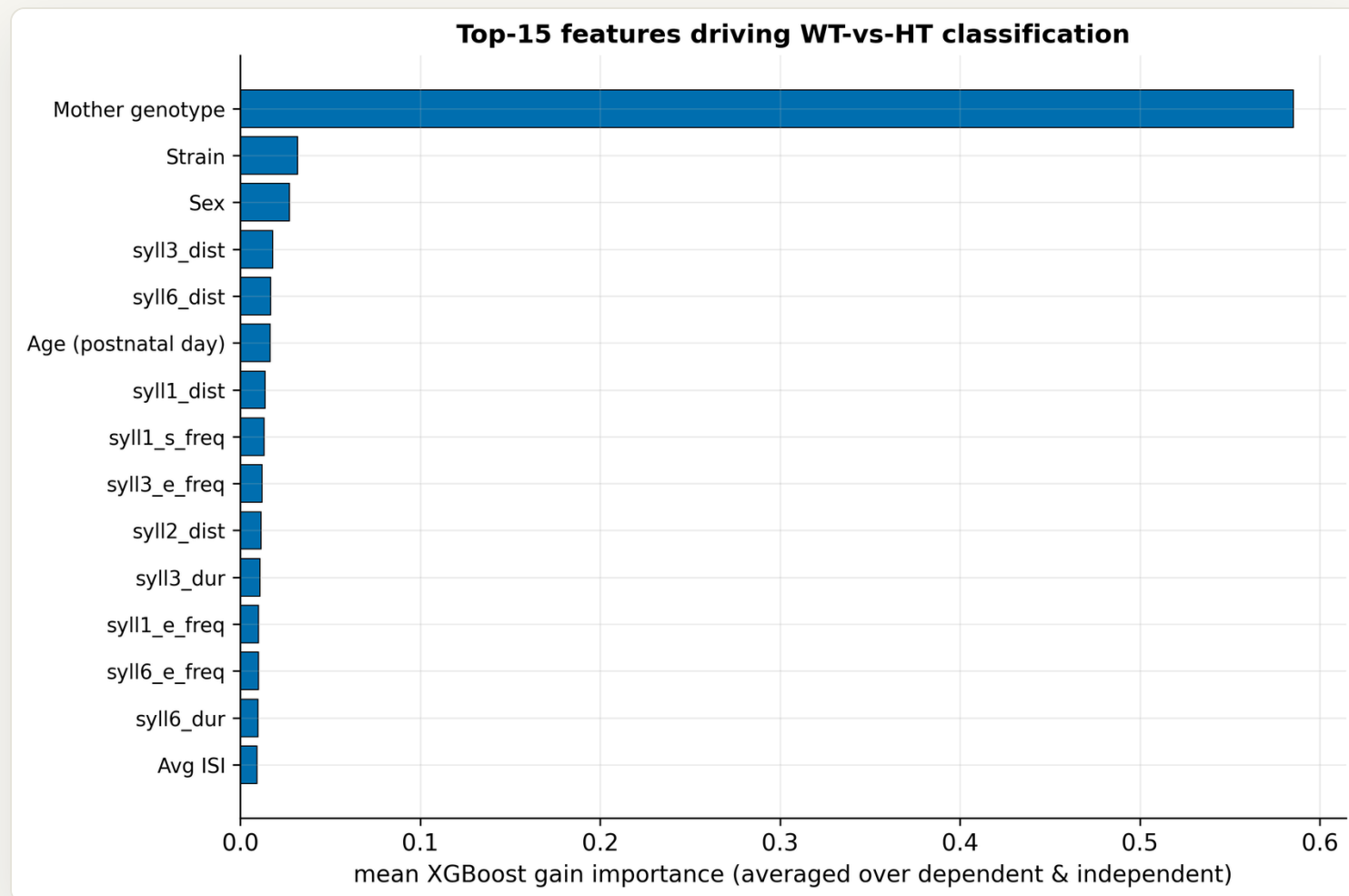
Questions welcome

Pipeline [github.com/AvielBitton/mouse-usv-asd-pipeline](https://github.com/AvielBitton/mouse-usv-asd-pipeline) App [Zenodo 10.5281/zenodo.20096810](https://zenodo.org/record/10.5281/zenodo.20096810)

## Master results — tabular models (pooled)

| MODEL               | SPLIT       | ACC         | WF1         | BAL ACC | ROC-AUC     | HT RECALL   | HT PREC | HT F1 |
|---------------------|-------------|-------------|-------------|---------|-------------|-------------|---------|-------|
| XGBoost (inherited) | dependent   | .733        | .749        | .795    | .876        | .940        | .496    | .649  |
| XGBoost (inherited) | independent | .693        | .706        | .678    | .770        | .637        | .452    | .529  |
| XGBoost-tuned       | dependent   | .772        | .785        | .798    | .885        | .844        | .543    | .661  |
| XGBoost-tuned       | independent | .702        | .719        | .725    | .753        | <b>.869</b> | .473    | .612  |
| TabPFN              | dependent   | <b>.781</b> | <b>.794</b> | .828    | <b>.908</b> | .918        | .550    | .688  |
| TabPFN              | independent | <b>.729</b> | <b>.743</b> | .662    | <b>.783</b> | .782        | .499    | .610  |

## Which features drive it — including a **confound**

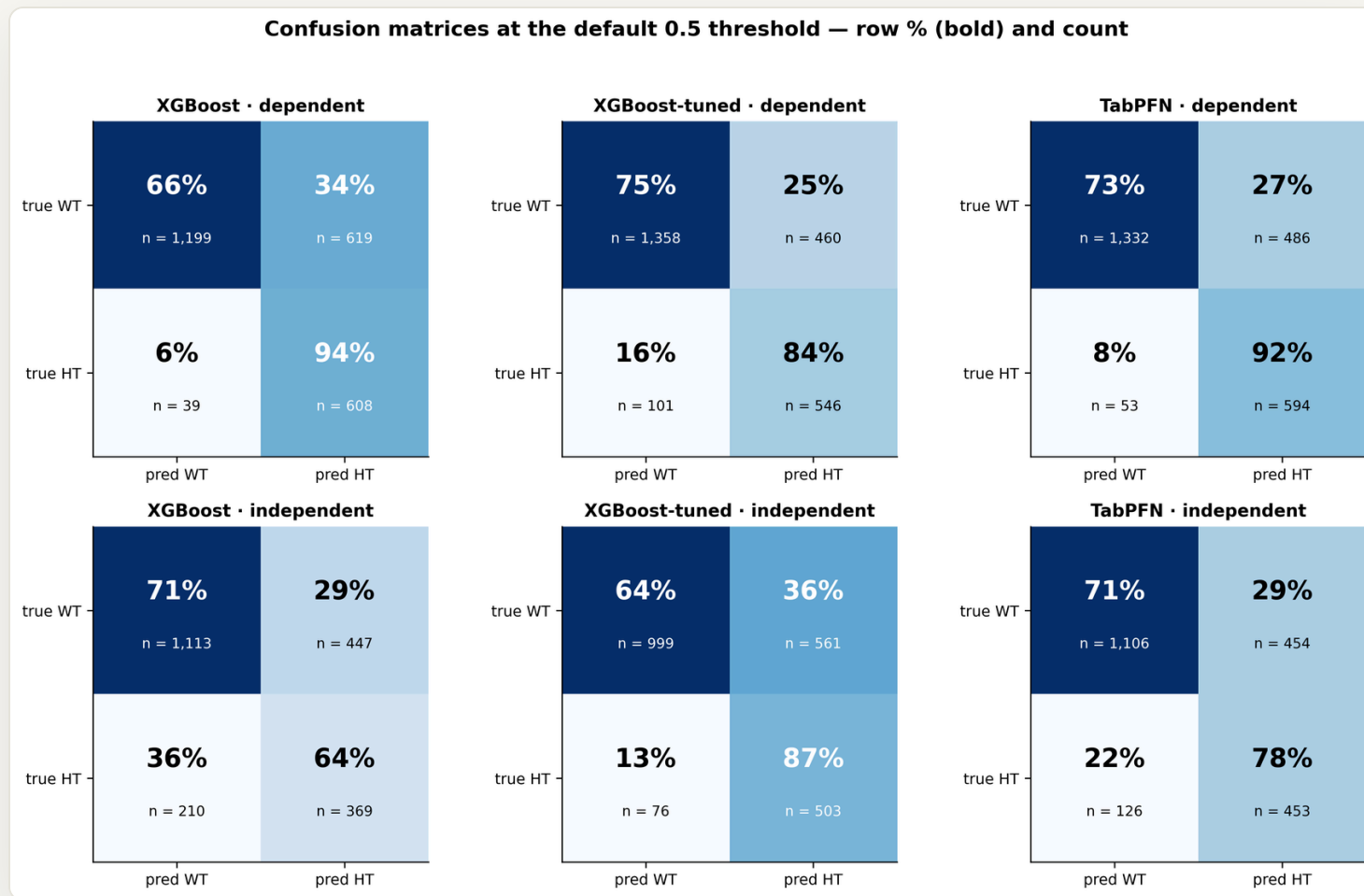


- ◆ **Mother genotype** tops gain importance — because it is predictive **by breeding**: a WT dam yields only WT pups.
- ◆ So the tabular model gets “free” stratification from **pedigree metadata**, not from the calls.
- ◆ Excluding metadata, **boundary frequencies & durations** lead — consistent with Shekel 2021.

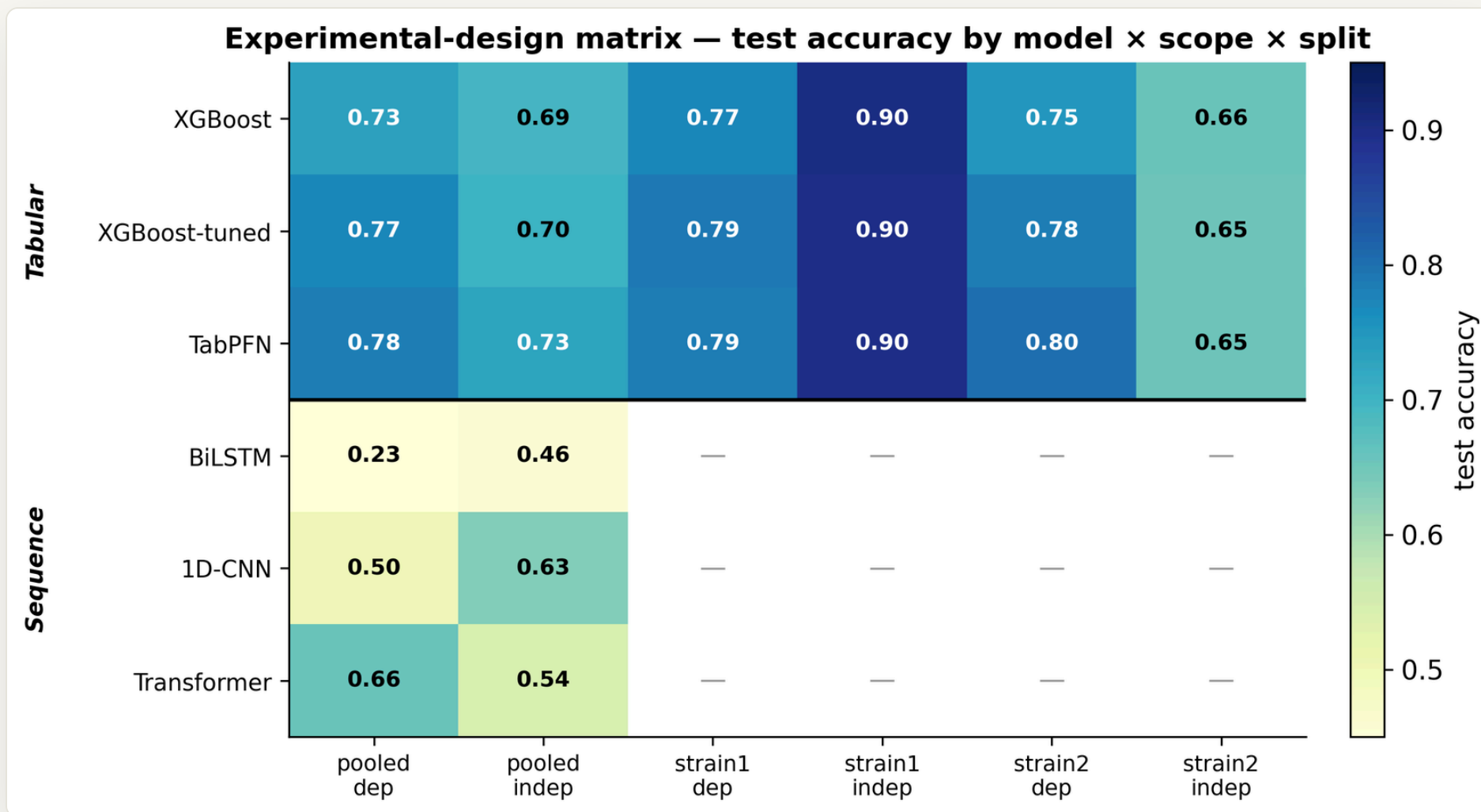
### WHY WE LEAN ON THE HARDER NUMBERS

The **subject-independent, AUC**, and **precision-wall** results reflect the genuine acoustic signal. A metadata-free ablation is future work.

# Confusion matrices — where the errors are



## What we ran — the design matrix



# Best model per scenario

| COHORT            | SPLIT       | BEST MODEL    | ACC  | WF1  | HT F1 |
|-------------------|-------------|---------------|------|------|-------|
| pooled            | dependent   | TabPFN        | .781 | .794 | .688  |
| pooled            | independent | TabPFN        | .729 | .743 | .610  |
| strain1           | dependent   | XGBoost-tuned | .789 | .802 | .649  |
| strain1           | independent | XGBoost       | .903 | .909 | .753  |
| strain2           | dependent   | TabPFN        | .801 | .809 | .703  |
| strain2           | independent | TabPFN        | .654 | .659 | .388  |
| pooled (sequence) | dependent   | Transformer   | .659 | —    | .481  |
| pooled (sequence) | independent | 1D-CNN        | .633 | —    | .267  |

## Decision-threshold tuning

### Within-subject

An F1 / target-recall objective lifts HT precision to **~0.62** (from .55) while holding recall ~0.81 — the only operating point that meaningfully clears the 0.50 wall.

### Across-subject

Youden / balanced objectives degenerate. A **target-recall** floor (~0.80) is the sensible screening choice: HT recall ~0.86 at precision ~0.46.

### The honest limit

Threshold tuning **relocates** the operating point on a fixed ROC — it **cannot** raise AUC or break the feature ceiling.